

EDUQUALITY ACADEMIC FOUNDATION

CENTER FOR ACADEMIC EXCELLENCE & UN SDG 4 QUALITY EDUCATION

"Quality Education – Learn Today, Build a Better Tomorrow."

PEDAGOGY & EDUCATIONAL TECHNOLOGY

Pedagogical Sciences & Educational Tech Toolkit

Cognitive Science of Learning, Instructional Systems Design, and Adaptive Assessment

Academic Information:

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Chapter 1: Cognitive Foundations of Learning

Section 1.0

This textbook has been compiled by university professors and pedagogical researchers to support deep conceptual comprehension and self-paced mastery. Use the structured review prompts at the end of each module.

CHAPTER 1

Cognitive Foundations of Learning

1.1 Working Memory, Cognitive Load Theory, and Schema Formation

Cognitive Load Theory identifies intrinsic, extraneous, and germane cognitive loads during learning. Working memory has limited bandwidth (Miller's 7 +/- 2 chunks); effective instructional design minimizes extraneous cognitive friction and scaffolds complex problem sets to foster long-term memory schema automation.

Key Conceptual Takeaways:

- Dual coding theory demonstrates superior retention when visual and verbal channels are harmonized.
- Spaced repetition and retrieval practice dramatically outperform massed passive cramming.

Selected Bibliography & Recommended Literature

- [1] UNESCO (2025). "Global Education Monitoring Report: Inclusion and Quality Benchmarks."
- [2] Goodfellow, I., Bengio, Y., & Courville, A. (2016). "Deep Learning." MIT Press.
- [3] Deisenroth, M. P., Faisal, A. A., & Ong, C. S. (2020). "Mathematics for Machine Learning." Cambridge University Press.
- [4] United Nations Sustainable Development Goals Knowledge Platform (2026). "Goal 4 Targets and Indicators."
- [5] Raworth, K. (2017). "Doughnut Economics: Seven Ways to Think Like a 21st-Century Economist." Chelsea Green Publishing.

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